

BIN381 Project

– Milestone 1

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1. Background and Stakeholder Context

South Africa continues to experience high levels of structural unemployment and persistent inequality with regards to the access of digital infrastructure. Research on South Africa's unemployed youth note that finding a job is becoming increasingly more reliant on digital communication technologies (Oyedemi & Choung, 2020). The digital divide caused by connectivity issues, lack of digital access, limited digital skills and high data costs limits the access of online job search applications for many jobseekers in South Africa. (Ohei & Alao, 2019; Shaw & Wheeler, 2023).

There have been Government initiatives such as South Africa Connect to try and close the gap, but digital access disparities persist and this was especially a problem during the COVID-19 pandemic (Department of Communications, 2013; Frans & Pather, 2022). Research on this topic regularly treats digital access as a contributor to employment outcomes, however another important question is whether digital access influences education. Education greatly effects employment outcomes and if people that have digital access improve their educational prowess, it would mean that digital access has a direct and indirect contribution to employment outcomes.

This hypothesis is important for resource allocation. If the relation between digital access and education is independent then they can be funded and evaluated separately, however if connectivity moderates education's effect on employment, then current intervention to improve education could be underperforming in areas without digital access. Connectivity investments can then be justified and could show returns on the existing spending on education rather than competing for the budget.

Stakeholder: The intended stakeholder is the National Treasury, who is responsible for advising the government on the allocation of development funding. It needs to decide how to prioritise limited intervention budgets between digital access subsidies and continued investment in education programmes. Their information need is reliant on whether these two categories of intervention are complements or substitutes, so that funding decisions reflect actual labour market dynamics rather than an assumed separation between the two policy areas.

2. Business Objectives and Success Criteria

Business Objective:

This project aims to determine whether household internet access improves education and employment outcomes in South Africa, and if so, how strongly. Access to the internet has increasingly become linked to opportunity, from finding job listings and applying online, to accessing learning material and staying connected during a job search. However, it remains unclear whether internet access strengthens the benefits of education. This matters for policy, since government spending is limited, and understanding this relationship can help determine whether digital access should be prioritised alongside continued investment in education.

Research suggests two possible ways internet access could relate to education and employment outcomes. In one pattern, having internet access makes education even more valuable, so someone who is both educated and connected does better than either factor would explain on its own. In the other pattern, internet access and education act more like substitutes, meaning one can make up for the lack of the other, so having internet matters less if someone is already well-educated. International research on education and employment support (Plavgo, 2023) shows that either pattern can occur, and which one shows up usually depends on the country and the specific group being studied, younger jobseekers, for example, tend to show the first pattern more often. Since this project focuses on the general working-age population in South Africa rather than youth specifically, it makes more sense to test which pattern applies here rather than assume it matches what other countries have found.

Business Success Criterion:

The project will be considered successful if it produces a clear, evidence-based answer as to whether household internet access is associated with better education and employment outcomes in South Africa, and whether this relationship strengthens the effect of education specifically. This includes determining whether people with both higher education and internet access show better employment outcomes than either factor would predict on its own, and whether internet access is associated with higher income among those who report it. The analysis should be transparent about the limitations of the data, particularly the salary variable's high non-response rate, so any conclusion is presented as an initial finding rather than a definitive claim.

3. Inventory of Resources

3.1. Analysis and Software Resources

- **Power Bi:** Power Bi is used to turn raw data from multiple sources (in our case the socioeconomic datasets) into interactive reports charts and dashboards. R scripts can be integrated directly into Power BI workflows rather than executing standalone R files for the purpose of showcasing the analysis done in R-Studio in an interactive and more visually appealing environment.
- **R and R-Studio:** R is the statistical programming language and software environment used for statistical computing, data analysis and graphics (graphs and other visuals). RStudio is the Integrated Development Environment (IDE) that we use that makes working with R much easier and more organised. R processes calculations, runs statistics and creates graphs and we can create R files or R scripts that we can then use in other software tools like Power Bi or to easily share amongst our fellow BIN students and/or lecturers.
- **Microsoft Excel and CSV's:** Excel is a spreadsheet program that allows us to organize, calculate, analyse and visualise data and CSV is textfile based filetype with a unique twist that it stores all values and separates them using commas. All data/information of the socioeconomic datasets are stored on CSV/Excel datasheets, which are used as the data sources that analysis using R is done on and the KPI creation on Power Bi is based on.

3.2. Data Resources

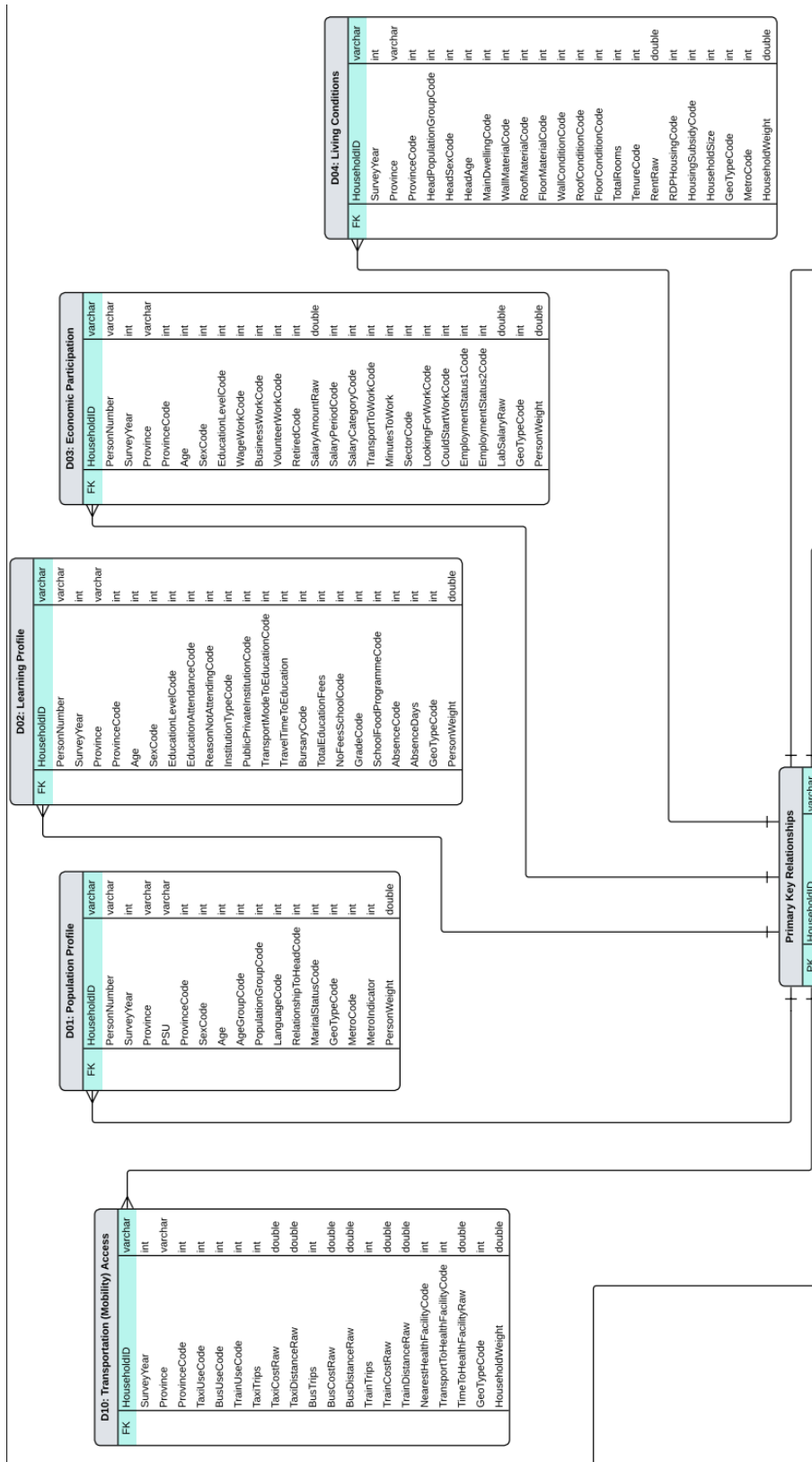
- **D01_Population_Profile.csv (Population and demographic profile)** - This dataset establishes who is in the population. This dataset is the demographic backbone every other person-level dataset depends on and it's what lets us break down any other finding by age, sex, population group, language, household role, etc. Without this dataset we don't know who is affected and how many in each household are affected by any other factors, such as those being described and stored in the other socioeconomic datasets.

- **D02_Learning_Profile.csv – (Education and learning profile)** - This dataset measures educational access and progression, whether people are in school, what's stopping them from attending school, how far they must travel to get there and whether cost or other factors form a barrier to education. It's the dataset we use to identify education-poverty pathways and can be used in a wide variety of scenarios that can be analysed for different purposes.
- **D03_Economic_Participation.csv (Economic participation)** - This dataset measures individual labour market engagement, not just whether someone works, but the nature of that work, what they earn, how far they commute and whether non-workers are actively job-seeking or have given up.
- **D04_Dwelling_Conditions.csv (Dwelling and housing conditions)** - This dataset captures physical housing adequacy, structure quality, overcrowding and tenure security. This dataset can be used as a starting point for a deprivation index, since poor dwelling quality correlates strongly with almost every other hardship measure in the other datasets.
- **D05_Basic_Services.csv (Water, sanitation and basic services)** - This dataset measures access to safe water and sanitation sources, reliability, perceived safety and toilet type/sharing. This is a core service delivery indicator where problems that occur are often geographically concentrated.
- **D06_Household_Energy.csv (Household energy)** - This dataset captures energy access and reliability, whether a household has electricity, cooking/lighting/heating fuel and load-shedding exposure. This dataset is useful for distinguishing connected, but unreliable households (load-shedding outages) from never connected households.
- **D07_Household_Wellbeing.csv (Food security and household well-being)** - This is the food security dataset that stores data about hunger, worries about running out of food, skipped meals and income.
- **D08_Connectivity_Access.csv (Connectivity and digital access)** - This dataset measures the populations digital access, such as if people have access to phones/internet and specifically whether people have internet for study or work purposes.

- **D09_Household_Resources.csv (Income, grants, assets and resource)** - This dataset is the income and asset dataset. It stores income sources, self-assessed income adequacy and a long asset inventory. This is the best dataset for household socio-economic status/wealth calculation, because there is not a clean income figure, however asset ownership is often more reliable than self-reported income for socioeconomic surveys, especially in South-Africa.
- **D10_Mobility_Access.csv (Mobility and access)** - This dataset measures transport access and geographic isolation, public transport use/cost/distance and distance/time to the nearest health facility.
- **D11_Labour_Market.csv (Labour-market modelling data)** - This dataset goes into a deeper, quarterly employment-quality overview of factors, such as unemployment duration, underemployment, informality, NEET status, occupation and industry.

3.3. ERD of All Datasets

(Note for the lecturer: We set up an ERD to showcase all the datasets and their shared attributes, not for indicating direct relationships, because there is no direct starting point and there are differences between their candidate keys)



...	int
WallConditionCode	int
RoofConditionCode	int
FloorConditionCode	int
TotalRooms	int
TenureCode	double
RentRaw	int
IDPHousingCode	int
HousingSubsidyCode	int
HouseholdSize	int
GeoTypeCode	int
MetroCode	int
HouseholdWeight	double

D05: Basic Services	
FK	HouseholdID
SurveyYear	int
Province	varchar
ProvinceCode	int
DrinkingWaterSourceCode	int
DistanceToWaterRaw	double
TimeToWaterRaw	int
PipedWaterCode	int
WaterSafeCode	int
WaterCleanCode	int
WaterTasteCode	int
WaterSmellCode	int
WaterTreatmentCode	int
MunicipalWaterSourceCode	int
WaterMeterCode	int
PaysForWaterCode	int
WaterInterruptions12MonthsCode	int
WaterInterruptionFrequencyCode	int
WaterAlternativeSourceCode	int
MainToiletCode	int
SharedToiletCode	int
ToiletLocationCode	int
ToiletDistanceCode	int
HandwashingFacilityCode	int
HandwashingCode	int
GeoTypeCode	int
HouseholdWeight	double

D06: Household Energy	
FK	HouseholdID
SurveyYear	int
Province	varchar
ProvinceCode	int
ElectricityAccessCode	int
MainElectricitySourceCode	int
MainElectricityCode	int
ElectricitySupplyCode	int
CookingEnergyCode	int
LightingEnergyCode	int
SpaceHeatingEnergyCode	int
LoadsheddingFrequencyCode	int
AlternativeLightingCode	int
AlternativeCookingCode	int
PaysForElectricityCode	int
SolarGeyserAssetCode	int
SolarPanelAssetCode	int
GeoTypeCode	int
HouseholdWeight	double

D07: Household Wellbeing	
FK	HouseholdID
SurveyYear	int
Province	varchar
ProvinceCode	int
AdultHungerCode	int
ChildHungerCode	int
WorriedFoodWouldRunOutCode	int
UnableToEatHealthyFoodCode	int
AteFewFoodsCode	int
FewFoodsDays	int
SkippedMealCode	int
SkippedMealDays	int
AteLessCode	int
AteLessDays	int
RanOutOfFoodCode	int
RanOutDays	int
HungryButDidNotEatCode	int
WholeDayWithoutEatingCode	int
HouseholdSize	double
TotalMonthlyHouseholdIncomeRaw	double
GeoTypeCode	int
HouseholdWeight	double

D08: Connectivity and Internet Access	
FK	HouseholdID
SurveyYear	int
Province	varchar
ProvinceCode	int
TelephoneCode	int
CellphoneCode	int
FixedInternetCode	int
MobileInternetCode	int
PublicWiFiInternetCode	int
StudyInternetCode	int
WorkInternetCode	int
LibraryInternetCode	int
CafeInternetCode	int
HouseholdMobilePhones	int
HouseholdSmartphones	int
ComputerAssetCode	int
GeoTypeCode	int
HouseholdWeight	double

D09: Household Resources	
FK	HouseholdID
SurveyYear	int
Province	varchar
ProvinceCode	int
IncomeSalaryCode	int
IncomeBusinessCode	int
IncomeRemittanceCode	int
IncomePensionCode	int
IncomeGrantCode	int
IncomeAgricultureCode	int
NonIncomeCode	int
OtherIncomeCode	int
MainIncomeSourceCode	int
RequiredIncomeRaw	double
ComparativeIncomeCode	int
ExpenditureBandCode	int
TotalMonthlyHouseholdIncomeRaw	double
VehicleAssetCode	int
TotalVehicles	int
TVAssetCode	int
PayTVAssetCode	int
ComputerAssetCode	int
WashingMachineAssetCode	int
FreezerAssetCode	int
FridgeAssetCode	int
MicrowaveAssetCode	int
GreyserAssetCode	int
SolarPanelAssetCode	int
RadioAssetCode	int
HouseholdSize	int
SalaryEarningsInHousehold	int
SocialGrantRecipients	int
TotalGrantRecipients	int
HouseholdWeight	double

D11: Labour Market	
FK	HouseholdID
PersonNumber	varchar
SurveyYear	int
Province	varchar
SurveyDateRaw	varchar
Quarter	int
ProvinceCode	int
GeoTypeCode	int
GenderCode	int
Age	int
PopulationGroupCode	int
MaritalStatusCode	int
EducationLevelCode	int
FieldOfStudyCode	double
EducationAttendanceCode	int
EmploymentStatusCode	int
ExpandedEmploymentStatusCode	int
UnemploymentStatusCode	double
EducationStatusCode	int
NEETCode	int
GraduateStatusCode	int
AgeGroupCode	int
OccupationCode	int
IndustryCode	int
SectorCode	int
Sector2Code	int
InformalEmploymentCode	int
LongTermUnemploymentCode	double
UnderemploymentCode	double
HoursWorkedRaw	double
MonthlyAmountRaw	double
QuarterWeight	double
AnnualWeight	double

Primary Key Relationships	
PK	HouseholdID

4. Risks, Assumptions, and Constraints

Before we identify the Risks, Assumptions and Constraints it is important that we differentiate between them first, because in some cases, depending on how you view the problem, they can be mixed, and it is important that we understand the differences between each one of them.

Terms	Definition
Risks	Something that could happen and would hurt the project if it does happen. Risks can be measured from less important to extremely important depending on their likelihood and impact.
Assumptions	Something believed to be true, but without any proof that it is true. You assume something is correct without verifying if it is correct, which can cause problems later.
Constraints	A fixed boundary condition that is already known to be true or that is confirmed to influence the validity of the outcome were it to be ignored.

4.1. Risks

Risk	Likelihood	Severity
1. HouseholdID gets rounded, because it is an 18-digit integer which makes R think all HouseholdID's are duplicates.	High	Severe
2. Codes in attributes may not mean what a person might think, a data dictionary is required.	Medium	High
3. Non-random ("structural") missingness mistaken for random missingness. Many sentinel-coded fields are missing because a skip pattern made the question inapplicable, not because collection failed.	Medium	High if imputed
4. Not all HouseholdID's refer to the same household in all datasets, meaning that if you create a link between the households in the different datasets and they don't refer to the same household the analysis becomes inaccurate.	Low	Medium
5. If certain attributes aren't combined, they can skew the results. For example, the PSU field and four distinct weight variables signal a stratified, multi-stage cluster sample, not a simple random sample.	Medium	Medium/High
6. Scope creep can occur, because there are so many datasets. The risk of over-combining variables into a model that is statistically busy but conceptually incoherent.	Medium	Medium

4.2. Assumptions

Assumption	Why the project depends on it
1. D01–D10 come from one survey and is structurally consistent with Stats SA's General Household Survey (GHS) and D11 from comes from a separate survey which is structurally consistent with the Quarterly Labour Force Survey (QLFS).	Determines whether D11 can be joined to D01–D10 at all and at what level (unit-record vs aggregate).
2. The household/person fields are broadly representative of South Africa in 2024. Sampling and weighting, if applied correctly, produce valid national/provincial estimates.	Any claim like “X% of households experience Y” relies on this.
3. Province/geography codes (ProvinceCode, GeoTypeCode, MetroCode) use the same coding scheme across all 11 datasets.	Needed for any geography-based join or comparison across datasets.
4. A record represents one real household/person as intended, no duplicate enumeration or fabricated/test rows in the export, except when time is involved such as in D11 with the quarterly calculations.	Underpins basic counting and rate calculations.
5. The 2024 reference period is comparable across datasets, D01–D10's single annual snapshot and D11's four quarterly rounds can be meaningfully related to the same calendar year.	Needed to combine, for example a Q2 labour-market condition (D11) with an annual dwelling condition (D04) as if describing the same year.

4.3. Constraints

Constraint	Implication for the project
1. Person-level files (D01, D02, D03, D11) and household-level files (D04–D10) sit at different units of analysis.	Combining across the two groups requires an explicit aggregation decision, this choice needs to be stated and justified.
2. D11 cannot be linked to D01–D10 at unit-record level.	Any pathway bringing in labour-market dynamics (D11) alongside GHS-style variables (D04–D10) must operate at an aggregated level (e.g. province $\times$ quarter), not a row-level merge.
3. No codebook or data dictionary was provided.	Time must be budgeted specifically to source and validate Stats SA's official value labels for GHS 2024 / QLFS 2024 before variables are interpreted or presented.
4. The project brief requires combining at least 3 of the 11 datasets to acts as a socio-economic pathway.	Directly shapes which risks/assumptions above are relevant.
5. Data is cross-sectional, not a multi-year time series. D01-D10 is a single 2024 snapshot and D11 spans only the four 2024 quarters.	No multi-year trend analysis is possible from this data alone.
6. De-identified microdata still carries statistical disclosure risk at fine geographic/demographic cross-tabulations.	Avoid publishing small cell counts/percentages at the finest geography level in the final report. We need to aggregate before publishing where cell sizes are small.
7. We need to stay within the constraints set by the original publishers, their usage terms and conditions. We break ethical boundaries if we do not.	Verify the usage terms disclosed by the original publishers to ensure no violations occur, which can cause legal problems later on.

5. Data-mining goals & technical success criteria

Analytics goal: Model employment status based on education level, digital access/connectivity, and the interaction between the two. The model will analyse for relevant demographic and geographic factors, such as age and urban/rural location.

Technical success criteria:

- The interaction term between education and connectivity is tested for statistical significance at $\alpha = 0.05$.
- The direction of any interaction effect, such as reinforcing vs substitution-compensation, is interpretable and reported as an odds ratio or equivalent effect size, following the interpretive convention used in comparable policy-complementarity research.
- Model achieves acceptable discriminative performance on a held-out test set.
- Findings are checked for robustness across at least two modelling approaches, such as logistic regression and a tree-based method.

6. Data Description

Stored in the BIN381 Project Data Dictionary Word Document in the Project Zip folder. The Data Description is too large for our main project Document so we added it separately.

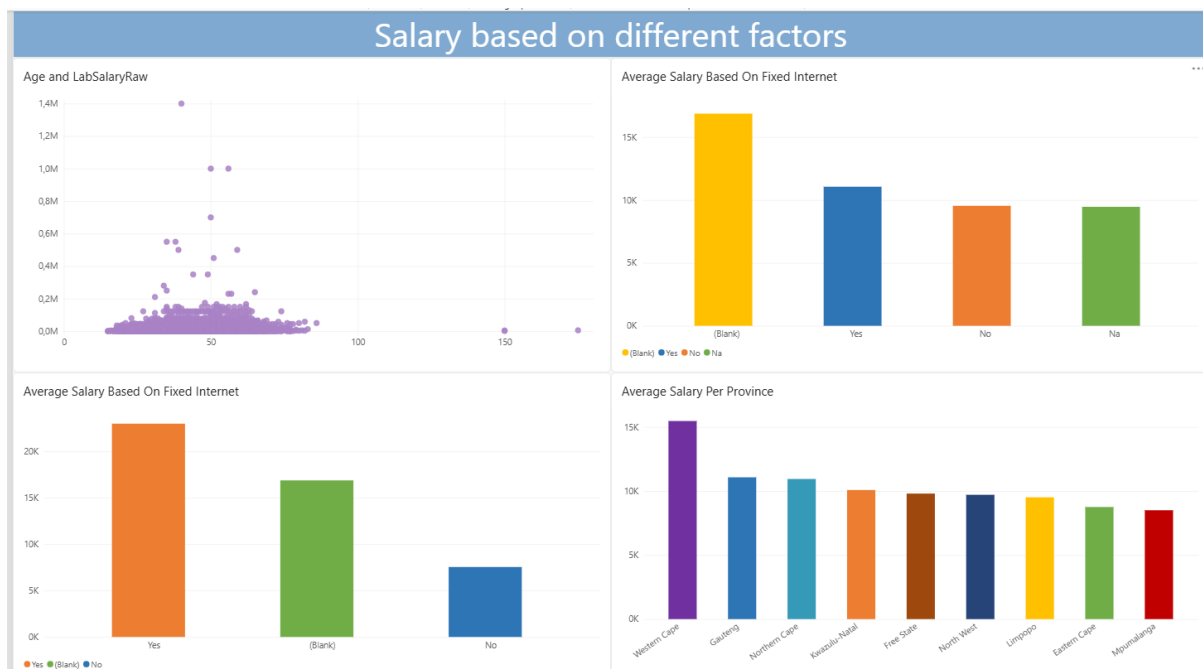
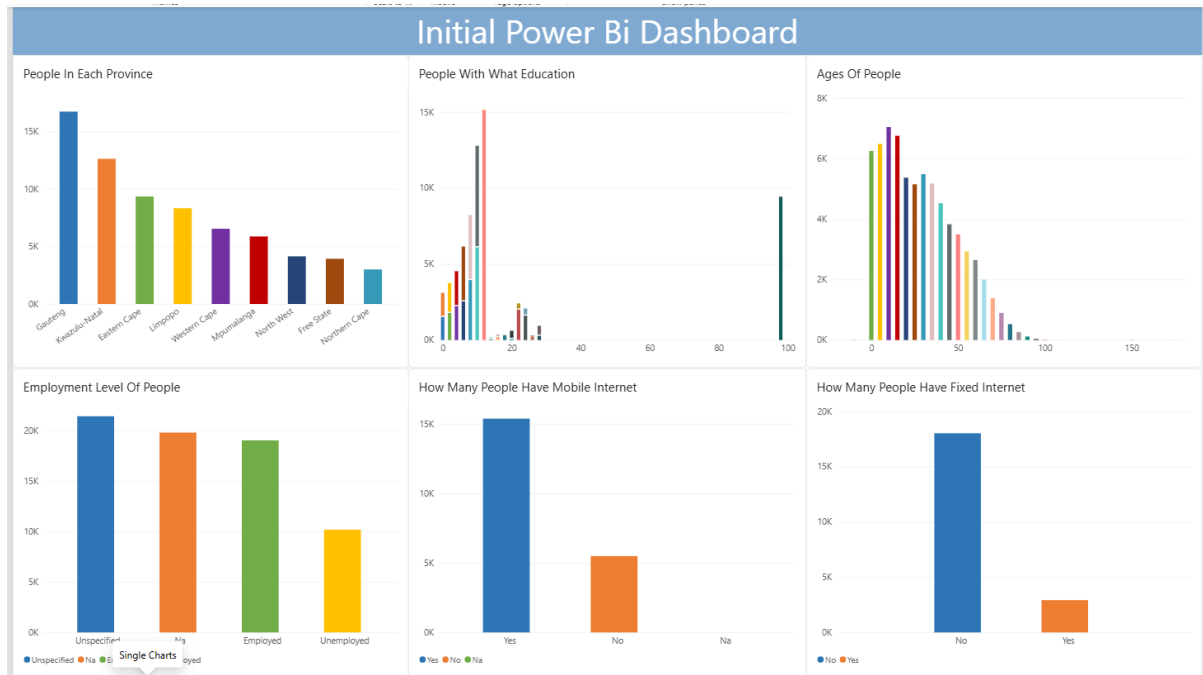
7. Data Exploration and Visualization

The EDA script and the Power Bi files can both be found in the project zip folder.

Here is the link to the github containing the Power Bi and the EDA script as well:

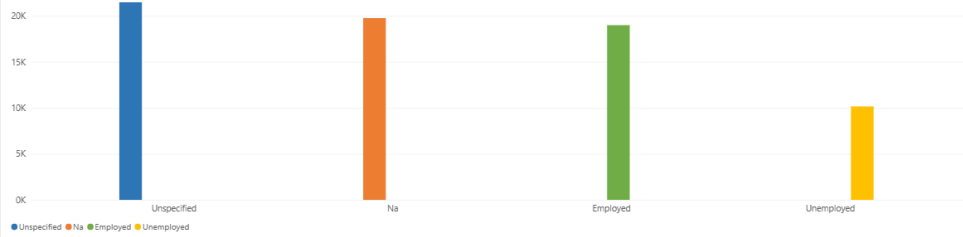
<https://github.com/Henri-Claassen/BIN381>

Images of the power Bi Are found Below:



Employment and Education Affected By Internet

Employment Status Of People



Slider

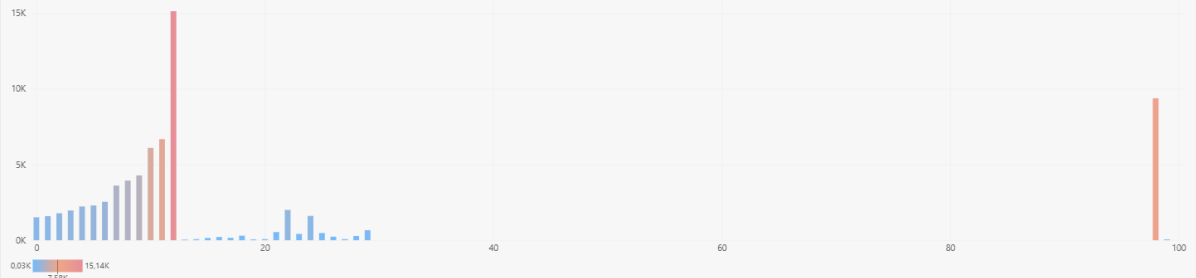
FixedInternetLabel

☐ (Blank)

☐ No

☐ Yes

Education Level Of People



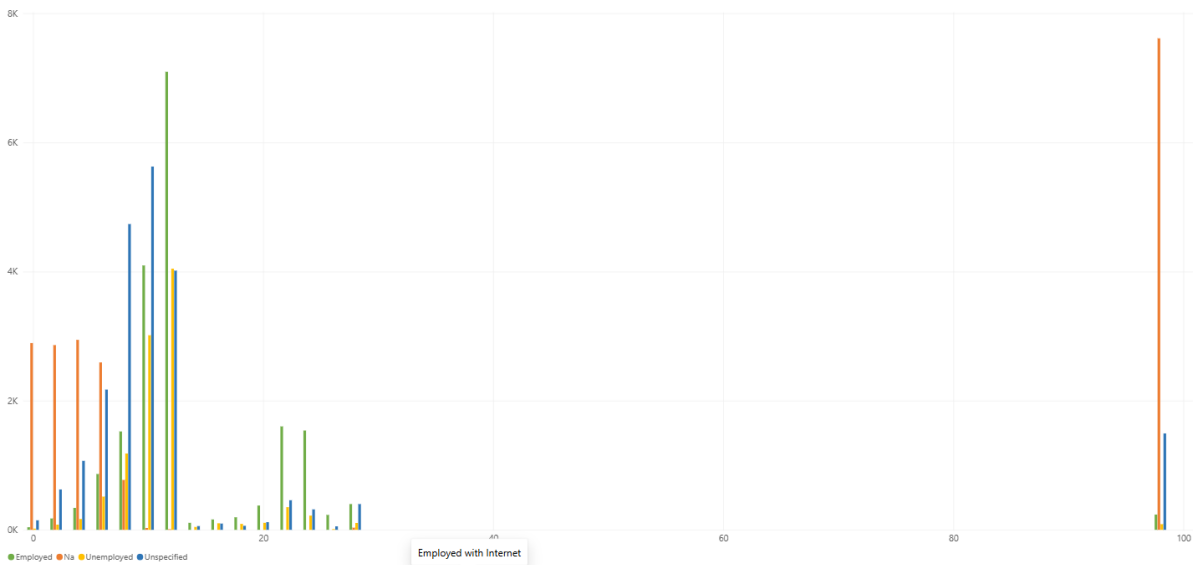
Education and Employment Based On Fixed Internet

FixedInternetLabel

☐ No

☐ Yes

Employment Based On Education



8. Data Quality Assessment

8.1. Purpose

This section consolidates the data quality assessment carried out on every dataset during Milestone 1. Four datasets were selected for the project: D01 (Population Profile), D02 (Learning Profile), D03 (Economic Participation), and D08 (Connectivity Access). It also documents the evidence supporting the exclusion of D11 (Labour Market). The R EDA notebook (EDA.qmd) was used for our findings and confirmed during initial Power BI exploration.

No dataset was assumed to be clean or ready for analysis. Each dataset was checked for completeness, uniqueness, validity, consistency, and structural integrity, following the assessment criteria set out in the project brief.

8.2. Methodology

The following checks were performed on each selected dataset:

- **Completeness:** count of missing (NA) values per column.
- **Uniqueness:** duplicate rows checked on the relevant key (a composite key for person level data, household_id for household-level data).
- **Validity:** values checked against the documented valid category codes, with numeric fields checked for impossible values (e.g. negative or implausible ages).
- **Consistency:** text fields checked for inconsistent representation of the same real-world value (e.g. capitalisation).
- **Structural integrity:** the unit of observation, key identification, and cross dataset relationships verified using intersect() on household_id and the composite keys.

8.3. Cross Cutting Issue: HouseholdID Precision

Across all the datasets, HouseholdID values are 17-18 digits long. R is only precise up to about 15-16 digits (2^{53}) by default since it uses the double numeric type. This resulted in read_csv()'s type detection automatically rounding the ID on import, corrupting them before analysis began. In the initial glimpse() output this was identified because of the ID being shown in scientific notation. This was then confirmed to have caused precision loss when looking for duplicates.

Resolution: Since the IDs were not required to be numeric all datasets were re-imported with HouseholdID explicitly as characters (`col_character()`) to keep the exact values. This must be consistently applied in any future re-import during upcoming milestones.

8.4. Dataset-by-Dataset Findings

D01 - Population Profile

Completeness

- household_id: 25 missing values
- age: 90 missing values

Uniqueness

23 duplicate rows were found on the household_id and person_number composite key. These appear linked to the missing household_id values noted above (rows with NA household_id may be grouping together as false duplicates). This this needs to be confirmed in M2 rather than being treated as a separate duplication issue.

Validity

Category code checks (province_code, sex_code, age_group_code, population_group_code, language_code, relationship_to_head_code, marital_status_code, geo_type_code, metro_code, metro_indicator) returned no invalid values against documented ranges.

Negative age values were found via filter ($\text{age} < 0$) and require correction or removal.

New finding (Power BI): an age vs salary scatter plot revealed two records with implausible ages of 150 and 180. The original R age-quality check only tested for negative values and did not test the upper bound. This has been added as a required check for M2.

Consistency

Province had three casing variants per province (e.g. "gauteng", "Gauteng", "GAUTENG"), initially inflating the apparent category count from 9 to 27 in exploratory bar charts. A secondary issue, leading/trailing whitespace or non-printable characters was also found after the initial capitalization fix, requiring an additional Trim/Clean step. Both issues were resolved in the EDA notebook using `str_to_title()` (R) and Trim/Format (Power Query), but must be carried through consistently into any future re-import.

D02 - Learning Profile

Completeness

- education_level_code: 120 missing values

Uniqueness

44 duplicate rows found on the household_id and person_number composite key. The cause has not been confirmed yet and requires investigation into whether this reflects a genuine data capture error or a valid repeated measurement scenario.

Validity

Category code checks were run across all coded columns (education_level_code, education_attendance_code, reason_not_attending_code, institution_type_code, and others). education_level_code and education_attendance_code were explicitly inspected for out of range values via filter().

D03 - Economic Participation

Completeness

- employment_status1code: 100 missing values

Uniqueness

40 duplicate rows found on the household_id and person_number composite key, same pattern as D02.

Validity

Category code checks were run across all coded columns (wage_work_code, business_work_code, employment_status1code, employment_status2code, sector_code, and others).

Major finding: SalaryAmountRaw is effectively unusable

Per the data dictionary, SalaryAmountRaw should hold a currency value (documented valid range: 1-7,500,000) when a valid SalaryPeriodCode is present. However, even restricting to the 7,462 rows where a genuine period (weekly/monthly/annual) was selected, values were almost universally 1. Not consistent with any real salary figure. This field does not appear to be reliably populated in this data extract and is not recommended for use in modelling without further investigation into the source data.

Major finding: LabSalaryRaw placeholder contamination

LabSalaryRaw uses two documented placeholder codes: 8888888888 ("don't know/refused") and 9999999999 ("not applicable"). These placeholders were found in 52,145 of 70,460 rows, approximately 74% of the dataset. Only 18,315 rows (26%) contain a genuine salary value. After excluding placeholders, the remaining distribution is right skewed (median of R5,257), which matches the expected income data shape once visualized on a log scale.

Implication: any analysis involving salary is working with a substantially reduced and potentially non-representative subset of respondents (those who reported income are not necessarily representative of the full sample). This should be explicitly flagged wherever LabSalaryRaw is used in later milestones.

The exact time period (weekly/monthly/annual) that LabSalaryRaw represents is not explicitly documented. Based on the magnitude of values (median of R5,257), this is inferred to most plausibly represent a monthly figure, consistent with typical South African income survey conventions, but this has not been confirmed against the original StatsSA source documentation.

D08 - Connectivity Access

Completeness

- mobile_internet_code: 60 missing values

Validity

Category code checks were run across all coded columns (telephone_code, cellphone_code, fixed_internet_code, mobile_internet_code, public_wi-fi_internet_code, study_internet_code, work_internet_code, library_internet_code, cafe_internet_code, household_mobile_phones, household_smartphones, computer_asset_code, geo_type_code).

Excluded Dataset: D11 - Labour Market

Quality issue identified

45 rows in hours_worked_raw contained inconsistent non-numeric placeholder text ("not recorded", "Unknown", "N/A") instead of a numeric value, causing a parsing warning on import. This reflects inconsistent data capture conventions for missing values within the same column.

Linkage issue: primary reason for exclusion

A key-relationship check (intersect() on household_id) between D01 and D11 returned zero overlapping values, out of a possible ~20,940 unique households in D01. This conclusively confirms D11 uses a separate identifier system from D01/D02/D03/D08, consistent with the project brief's note that D11 originates from a separate survey environment. D11 cannot be reliably linked to the other selected datasets at either the household or person level.

Decision: D03 (Economic Participation) was used as the source for employment outcome data instead, since it shares near-identical row counts with D01 (70,460 vs 70,440) and passed both household level and person level composite key verification.

8.5. Conclusion

The four selected datasets (D01, D02, D03, D08) are structurally linkable, with strong household and person level match rates. This supports the project's chosen analytical design. However, several genuine data quality issues were found. The most significant are SalaryAmountRaw being nearly completely unreliable and LabSalaryRaw having a high placeholder rate, both of which substantially affect what can be confidently concluded from income related analysis. These issues, along with the smaller completeness and duplication concerns across all four datasets are catalogued above for to be resolved in Milestone 2.

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